

The phrase "Big Data Loop" refers to

- ☐ a kind of big data processing framework which processes big data using a sequence of phases
- ☒ the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites 
- ☐ an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:

the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

The phrase "Big Data Loop" refers to

- ☐ a kind of big data processing framework which processes big data using a sequence of phases
- ☒ a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data ✓
- ☐ an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:

a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data

In a Chi-square test for feature selection of a term t with respect to a class c , the p-value is 0.015

- ☒ a. the term t is not selected as a feature if the confidence level is 0.99
- ☐ b. the term t is selected as a feature if the confidence level is 0.99
- ☐ c. the term t is not selected as a feature if the confidence level is 0.95
- ☐ d. the term t is selected as a feature if the confidence level is larger than 0.99



Your answer is correct.

La risposta corretta è:

the term t is not selected as a feature if the confidence level is 0.99

In a Chi-square test for feature selection of a term t with respect to a class c , the p-value is 0.015

- ☐ a. the term t is selected as a feature if the confidence level is 0.98
- ☐ b. the term t is selected as a feature if the confidence level is 0.99
- ☒ c. the term t is not selected as a feature if the confidence level is 0.95
- ☐ d. the term t is selected as a feature if the confidence level is larger than 0.99



Your answer is incorrect.

The correct answer is:

the term t is selected as a feature if the confidence level is 0.98

In the Hadoop system, the MapReduce application client

- ☐ centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- ☐ autonomously computes the resources each worker is allowed to use
- ☒ sends requests to YARN for resource allocation



Your answer is correct.

La risposta corretta è:

sends requests to YARN for resource allocation

In Spark's structured API, if Salary is a column of a data frame Emp, then Salary's values can be displayed by executing

- ☒ a. `from pyspark.sql.functions import column
Emp.select('Salary').show()` ✓
- ☐ b. `from pyspark.sql.functions import column
Emp.columns.select('Salary').show()`
- ☐ c. `from pyspark.sql.functions import column
Emp.column('Salary').show()`

Your answer is correct.

La risposta corretta è:

```
from pyspark.sql.functions import column  
Emp.select('Salary').show()
```

Which of the following is the correct contrastive loss ?

($d(v, w)$ denotes the euclidean distance between v and w ; x_1, x_2 are the embeddings of two inputs and y_1 and y_2 their two classes respectively, α is the margin)

- ☐ a. $L = d(x_1, x_2)$ if $y_1 == y_2$
 $L = \max(0, \alpha - d(x_1, x_2))$ if $y_1 \neq y_2$
- ☐ b. $L = d(x_1, x_2)$ if $y_1 \neq y_2$
 $L = \max(0, \alpha - d(x_1, x_2))$ if $y_1 == y_2$
- ☒ c. $L = d(x_1, x_2)$ if $y_1 \neq y_2$
 $L = \max(0, \alpha + d(x_1, x_2))$ if $y_1 == y_2$
- ☐ d. $L = d(x_1, x_2)$ if $y_1 == y_2$
 $L = \min(0, d(x_1, x_2) - \alpha)$ if $y_1 \neq y_2$



Your answer is incorrect.

The correct answer is:

$$L = d(x_1, x_2) \text{ if } y_1 == y_2$$
$$L = \max(0, \alpha - d(x_1, x_2)) \text{ if } y_1 \neq y_2$$

The function to create a decision tree classifier object in MLlib is

- ☒ a. `DecisionTreeClassifier()` ✓
- ☐ b. `DecisionTreeClassifier().transform()`
- ☐ c. `DecisionTreeClassifier().fit()`

Your answer is correct.

The correct answer is:

`DecisionTreeClassifier()`

In Spark's structured API, DataFrames and Datasets are data collections in the form of tables

- ☒ a. for which a schema is defined
- ☐ b. with columns of different lengths
- ☐ c. with untyped columns



Your answer is correct.

La risposta corretta è:

for which a schema is defined

In the structured API of Spark, the DataFrame class has a method where(). Executing

```
df.where(arg)
```

where df is DataFrame object, returns a DataFrame object

- ☒ with every row of df, except the ones that do not verify expression arg
- ☐ with exactly the rows of df which are also in the data frame passed as arg
- ☐ with exactly the rows of df equalling the row passed as arg



Your answer is correct.

La risposta corretta è:

with every row of df, except the ones that do not verify expression arg

If df is a Spark DataFrame, df.selectExpr() is a DataFrame method that

- ☐ chooses an expression to compute among the expressions passed to the method as arguments
- ☐ creates a data frame consisting of every row of df satisfying the expression passed to the method as argument
- ☒ creates a data frame such that each of its rows is computed as the values of the sequence of the expressions passed to the method as arguments ✓

Your answer is correct.

La risposta corretta è:

creates a data frame such that each of its rows is computed as the values of the sequence of the expressions passed to the method as arguments

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as possible. Which of the following is a sensible approach to address this task ?

- ☐ a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- ☐ b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)
- ☐ c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- ☒ d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird) ✗

Your answer is incorrect.

La risposta corretta è:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

The filter() method of DataFrame in Spark's structured API

- ☐ retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- ☐ selects specified, possibly renamed, columns of a data frame
- ☒ accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition 

Your answer is correct.

La risposta corretta è:

accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

The Fusion-in-Encoder method is

- ☐ a. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and this single combined input is forwarded to the encoder.
encoder receives and encodes the input instance, provided to the input layer, that is concatenated with k external items, for example retrieved by a neural layer, and the resulting vecntor is encoded
- ☐ b. a merging strategy where the input instance provided to the input layer is combined with k external items, for example retrieved by a dense passage retriever, and these k combined inputs are forwarded to the encoder.
- ☐ c. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and the resulting input is forwarded to the decoder.
- ☒ d. a merging strategy where the input instance provided to the input layer is concatenated with k external items,  for example retrieved by a dense passage retriever, and these k combined inputs are forwarded to the encoder and decoder.

Your answer is incorrect.

The correct answer is:

a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and this single combined input is forwarded to the encoder.

encoder receives and encodes the input instance, provided to the input layer, that is concatenated with k external items, for example retrieved by a neural layer, and the resulting vecntor is encoded

Domanda **7**

Parzialmente
corretta

Punteggio
ottenuto 0,50 su
1,00



Contrassegna
domanda

In the Hadoop Distributed File System, which of the following is a function of the **client**?

Scegli una o più alternative:

- ☐ a. Store the state of the filesystem
- ☐ b. Notify its state to other nodes
- ☐ c. Deliver the data
- ☒ d. Execute the user code ✓
- ☐ e. Initiate the access to the distributed file system
- ☐ f. Track the ownership of the data

Risposta parzialmente esatta.

Hai selezionato correttamente 1.

Le risposte corrette sono: Execute the user code, Initiate the access to the distributed file system

Domanda **3**

Risposta
corretta

Punteggio
ottenuto 1,00 su
1,00

Contrassegna
domanda

In the Hadoop Distributed File System, which of the following is a function of the **datanode**?

Scegli una o più alternative:

- ☒ a. Notify its state to other nodes ✓
- ☐ b. Track the ownership of the data
- ☐ c. Execute the user code
- ☐ d. Initiate the access to the distributed file system
- ☐ e. Store the state of the filesystem
- ☒ f. Deliver the data ✓

Risposta corretta.

Le risposte corrette sono: Deliver the data, Notify its state to other nodes

Domanda **1**

Parzialmente
corretta

Punteggio
ottenuto 0,60 su
1,00



Contrassegna
domanda

In the Hadoop Distributed File System, which of the following is a function of the **namenode**?

Scegli una o più alternative:

- ☒ a. Store the state of the filesystem ✓
- ☒ b. Initiate the access to the distributed file system ✗
- ☐ c. Execute the user code
- ☒ d. Notify its state to other nodes ✓
- ☐ e. Track the ownership of the data
- ☐ f. Deliver the data

Risposta parzialmente esatta.

Hai selezionato correttamente 2.

Le risposte corrette sono: Track the ownership of the data, Store the state of the filesystem, Notify its state to other nodes

High-Performance Computing (HPC) can perform distributed computation. However, it is hard to program an HPC application for big data processing because HPC

- ☒ a. requires programmers to control coordination explicitly in programs, using message-passing primitives ✓
- ☐ b. places data randomly on the nodes, with random overlaps
- ☐ c. is not flexible enough in handling node failures, as it handles them automatically

Your answer is correct.

La risposta corretta è:

requires programmers to control coordination explicitly in programs, using message-passing primitives

A difference between High performance computing (HPC) and MapReduce is

- ☐ the higher number of computing units which can be used in MapReduce with respect to HPC
- ☐ the maximum size of problems which can be solved using MapReduce
- ☒ the reduced amount of programming effort required to program an application in MapReduce with respect to HPC, if the problem admits a solution in terms MapReduce computing primitives ✓

Your answer is correct.

La risposta corretta è:

the reduced amount of programming effort required to program an application in MapReduce with respect to HPC, if the problem admits a solution in terms MapReduce computing primitives

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- ☐ creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- ☐ creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- ☒ creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df 

Your answer is correct.

La risposta corretta è:

creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

In Spark's structured API, to create a data frame named `df` from a JSON file named `f_name`, the following assignment can be used

- ☒ a. `df = spark.read.format("json").load(f_name)` ✓
- ☐ b. `df = spark.read.load(f_name)`
- ☐ c. `df = spark.format("json").load(f_name)`

Your answer is correct.

La risposta corretta è:

```
df = spark.read.format("json").load(f_name)
```

the latent semantic analysis allows to reduce

- ☐ a. the rank of the original term document matrix
- ☐ b. the dimensions of the original term document matrix
- ☐ c. the rank of the rows of the original term document matrix
- ☒ d. the rank of the columns of the original term document matrix

Your answer is incorrect.

The correct answer is:

the rank of the original term document matrix

The map function in MapReduce

- ☐ always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- ☐ is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- ☒ is a user defined function which is applied to every key-value pair of the dataset 

Your answer is correct.

Le risposte corrette sono:

always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

In the MapReduce computational model, how can the client control the **Shuffle** step?

Scegli un'alternativa:

- ☐ a. Check the status of the network and distribute the jobs appropriately
- ☐ b. Set the level of parallelism
- ☐ c. Set the maximum response time which can be accepted
- ☒ d. No control ✓ As a matter of fact, the shuffle is influenced by deployment parameters and by the status of the network, not by the client

Risposta corretta.

La risposta corretta è: No control



Domanda **6**

Risposta
corretta

Punteggio
ottenuto 1,00 su
1,00



Contrassegna
domanda

In the MapReduce computational model, which is the role of the **Map** step?

Scegli un'alternativa:

- ☐ a. Distribute to the *shuffler* the chunks of data which will be reduced and forwarded to the client
- ☐ b. To be aware of the status of the network and distribute the jobs appropriately
- ☒ c. Distribute to the *mappers* the chunks of data and the map function, in order to generate the key-value pairs which will be shuffled and passed to the reducers ✓
- ☐ d. Prepare the chunks of data to be forwarded to the reducers

Risposta corretta.

La risposta corretta è: Distribute to the *mappers* the chunks of data and the map function, in order to generate the key-value pairs which will be shuffled and passed to the reducers

In the MapReduce programming model

- ☐ key-value pairs are the input of the computation and values are the output of the computation
- ☐ the map function maps each value to a unique key according to a user supplied mapper
- ☐ key-value pairs are transformed into key-value pairs by a user-specified mapper

Your answer is incorrect.

The correct answer is:

key-value pairs are transformed into key-value pairs by a user-specified mapper

Domanda **9**

Risposta
corretta

Punteggio
ottenuto 1,00 su
1,00



Contrassegna
domanda

In the MapReduce computational model, which is the role of the **Reduce** step

Scegli un'alternativa:

- ☐ a. Return to the shuffler the results of the reduction
- ☒ b. Prepare the results for the delivery to the client ✓
- ☐ c. To be aware of the status of the network and distribute the jobs appropriately
- ☐ d. Distribute to the *mappers* the results of the computations

Risposta corretta.

La risposta corretta è: Prepare the results for the delivery to the client

In the MapReduce computational model, which is the role of the **Shuffle** step

Scegli una o più alternative:

- ☐ a. Prepare the results for the delivery to the client
- ☐ b. Return to the mappers the results of the reduction
- ☒ c. Distribute to the *reducers* the results of the computations executed by the *mappers* ✓
- ☐ d. To be aware of the status of the network and distribute the jobs appropriately

Which are the components of a general message-passing layer in a graph neural network?

- ☐ a. Message function, Permutation Invariant Aggregator, Differentiable Function
- ☐ b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- ☒ c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator
- ☐ d. Message function, Permutation Invariant Aggregator, Attention Mechanism



Your answer is incorrect.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

The outer join supported by the structured API of Spark

- ☒ may output rows with null values for all the columns of one of the joined tables
- ☐ always outputs null values for all the columns of the joined table having the least number of rows
- ☐ always outputs rows with null values for all the columns of at least one of the joined tables



Your answer is correct.

La risposta corretta è:

may output rows with null values for all the columns of one of the joined tables

In the structured API of Spark, `pyspark.sql.Row`

- ☐ is method of `DataFrameReader` which gets one row at a time from a JSON text file
- ☒ can be used to create an object which can be passed to a `spark.createDataFrame` method to add rows to a data frame 
- ☐ is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:

can be used to create an object which can be passed to a `spark.createDataFrame` method to add rows to a data frame

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$) ?

- ☒ a. The product between Q and K^T
- ☐ b. The product between $\text{softmax}(QK^T)$ and V
- ☐ c. The factor $1/\sqrt{d_k}$
- ☐ d. The V matrix



Your answer is correct.

La risposta corretta è:

The product between Q and K^T

The R-precision of a query q measures

- ☒ a. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q ✓
- ☐ b. the number of relevant documents for the query q in the first R retrieved docs, divided by the number of docs in the corpus
- ☐ c. the number of relevant documents for the query q in the first R retrieved docs, divided by R which is the number of docs retrieved
- ☐ d. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for each q

Your answer is correct.

La risposta corretta è:

the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q

Traditional Relational Data Base Management Systems (RDBMS) can perform distributed computation, but they are unsuitable for big data processing because

- ☒ a. their technology is optimised for short accesses to small amounts of data
- ☐ b. they are optimised for read-only access to large amounts of data
- ☐ c. they cannot operate on GB-sized data efficiently



Your answer is correct.

La risposta corretta è:

their technology is optimised for short accesses to small amounts of data

A difference between an RDBMS and MapReduce is

- ☐ that an RDBMS is capable of performing distributed computation
- ☒ that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses 
- ☐ that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:

that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

To read a CSV file into a data frame in Spark's Python API

- ☐ a. functions `format()` and `load()` can be used
- ☒ b. the `csv.reader()` function of Python's `csv` module must be called
- ☐ c. an `import csv` statement must be used



Your answer is incorrect.

The correct answer is:

functions `format()` and `load()` can be used

The select method of DataFrame, called on a data frame df,

- ☐ generates a data frame containing selected rows of df
- ☒ generates a data frame containing selected columns of df
- ☐ generates a data frame containing selected rows and columns of df



Your answer is correct.

La risposta corretta è:

generates a data frame containing selected columns of df

Stemming and Lemmatization

- ☒ a. are text processing techniques to cut suffixes and reduce each word to its root and base form respectively ✓
- ☐ b. are text processing techniques to cut suffixes and reduce each word to its base form and root respectively
- ☐ c. are text processing techniques to cut prefix and reduce each word to its base form and root respectively
- ☐ d. are text processing techniques to cut prefix and reduce each word to its root and base form respectively

Your answer is correct.

La risposta corretta è:

are text processing techniques to cut suffixes and reduce each word to its root and base form respectively

Why the TF-IDF of a stop word is zero ?

- ☐ a. because the IDF is 1
- ☐ b. because the TF is 1
- ☒ c. because the IDF is 0
- ☐ d. because the TF is 0



Your answer is incorrect.

La risposta corretta è:
because the IDF is 1

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

- ☒ a. 1. onehot encoding vector of q with respect to the dictionary ✔
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

- ☐ b. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

- ☐ c. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

- ☐ d. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

In Spark's MLlib, transform() is a function that

- ☒ can be used to compute predictions from a machine learning model
- ☐ standardises the dataset to have mean 0 and variance 1 in all its numeric columns
- ☐ is only used to compute predictions from a machine learning model



Your answer is correct.

La risposta corretta è:

can be used to compute predictions from a machine learning model

Which is the correct sequence of layers in a typical Transformer Encoder (without considering residual connections and normalization layers) ?

- ☒ a. Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward
- ☐ b. Feed Forward -> Input embeddings -> Position Embeddings -> MultiHead Attention
- ☐ c. Position Embeddings -> Input embeddings -> MultiHead Attention -> Feed Forward
- ☐ d. Input embeddings -> Position Embeddings -> Feed Forward -> MultiHead Attention



Your answer is correct.

La risposta corretta è:

Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward

Which of the following is the correct equation of the triplet loss ?

($d(x, y)$ denotes the euclidean distance between x and y ; a , p and n are the embeddings of the anchor, positive and negative sample, respectively)

- ☒ a. $L = \max(0, d(a, p) - d(a, n) + \alpha)$
- ☐ b. $L = \max(0, d(a, p) + d(a, n) + \alpha)$
- ☐ c. $L = \max(0, \alpha - d(a, p) + d(a, n))$
- ☐ d. $L = \max(0, \alpha - d(a, p) - d(a, n))$



Your answer is correct.

La risposta corretta è:

$$L = \max(0, d(a, p) - d(a, n) + \alpha)$$

the dimension of word embedding vectors

- ☐ a. is a hyperparameter decided by the data scientist
- ☒ b. corresponds to the vocabulary length
- ☐ c. corresponds to the number of documents
- ☐ d. is automatically defined by the neural network



Your answer is incorrect.

The correct answer is:

is a hyperparameter decided by the data scientist

In the Hadoop system, YARN

- ☒ a. fulfills application client requests to start an application using its resource manager to ask node managers to allocate resources ✓
- ☐ b. uses its resource manager to directly create containers
- ☐ c. runs one node manager and a varying number of resource managers in the cluster

Your answer is correct.

La risposta corretta è:

fulfills application client requests to start an application using its resource manager to ask node managers to allocate resources